

MAT 310, Handout 28: Normal Approximation to the Binomial Distribution (7.5)

Recall: Binomial Distribution

- $Y \sim \text{Bin}(n, p)$ models the number of successes in n independent trials, each with only two potential outcomes.
- p = probability of success
- $p(y) =$
- $E(Y) =$
- $V(Y) =$

Example 1. Suppose a basketball player makes 60% of his free throws. Assuming each shot is independent, find the probability that he makes at least 25 free throws if he shoots 50.

Normal Approximation to the Binomial Distribution

Let $Y \sim \text{Bin}(n, p)$. The RV Y can be viewed as the sum of a sample of 0s and 1s; that is, $Y = \sum_{i=1}^n X_i$, where $X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{o.w.} \end{cases}$. The X_i are independent (since the trials are independent) and are called *Bernoulli RVs*.

- $E(X_i) =$
- $V(X_i) =$
- $\frac{Y}{n} =$

Therefore, for large n ,

$$\frac{Y}{n} \dot{\sim}$$

$$Y \dot{\sim}$$

Question: When is the normal distribution a good approximation to the binomial distribution?

Rule of Thumb:

Remark: When approximating the probability of a binomial (discrete) random variable X with a normal (continuous) random variable W , we should use the continuity correction factor as follows:

- $P(X = x) \approx P(x - .5 < W < x + .5)$
- $P(X \geq x) = P(X > x - 1) \approx P(W > x - .5)$
- $P(X \leq x) = P(X < x + 1) \approx P(W < x + .5)$

Example 2. Consider the 60% free throw shooter from the previous example. If he shoots 50 free throws, find the probability (using the normal approximation) that he makes at least 25 of them.